

ASWATH RAMANA

Senior GenAI Engineer

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PROFESSIONAL SUMMARY

Senior GenAI Engineer and Technical Lead with 7+ years of experience delivering scalable enterprise AI/ML solutions across banking, insurance, and financial services. Specialized in architecting production-grade RAG pipelines, multi-agent orchestration (LangGraph, LangChain), text-to-SQL systems, and intelligent document processing. Proven track record leading cross-functional engineering teams through full lifecycle delivery on AWS and Azure—notably driving sanctions-screening automation that reduced false positives from 55% to 10% and deploying enterprise voice simulators serving 1,000+ users.

TECHNICAL SKILLS

- **LLMs & Providers:** Azure OpenAI, OpenAI GPT, Google Gemini, Groq API
- **Agents & Orchestration:** LangGraph, LangChain, multi-agent workflows, state management, tool integration, structured outputs, schema validation, prompt engineering, human-in-the-loop escalation
- **Retrieval & Documents:** RAG, Azure AI Search, FAISS, embeddings, vector search, text-to-SQL, Tesseract OCR
- **ML & NLP:** Scikit-learn, spaCy, NLTK, entity resolution, token classification, anomaly detection, fuzzy and phonetic matching
- **Backend & Databases:** Python, FastAPI, Flask, REST APIs, SQL, PostgreSQL, SQL Server, Supabase
- **Cloud & Enterprise Integration:** AWS, Amazon S3, Azure AI Foundry, Azure Speech Services, Azure Bot Service, Azure Functions, Azure Logic Apps, SharePoint, Microsoft Dynamics 365, Power Automate
- **Deployment & Quality:** Git, GitLab, CloudBees CI/CD, Azure DevOps, Docker, APScheduler, guardrails, LLM evaluation, benchmarking, regression testing
- **Reporting:** Power BI, Tableau

PROFESSIONAL EXPERIENCE

Senior GenAI Consultant | EY GDS | November 2025 – Present | Chennai, India

- Lead 5–6 developers and mentor junior engineers on RAG microservices and LangGraph agent orchestration for business intelligence and insurance applications.
- Own architecture, regression testing, and AWS CI/CD releases via GitLab and CloudBees, delivering a BI assistant and an insurance voice simulator used by 1,000+ employees.

Senior Software Engineer | Qentelli Solutions | June 2024 – October 2025 | Hyderabad, India

- Led data science and engineering teams of 3–5 to design and build multi-agent finance assistants, text-to-SQL/RAG applications, and OCR-based workflows.
- Owned backend architecture, APIs, database schemas, and LLM orchestration workflows for client-funded POCs and enterprise applications.

Senior Consultant | Ernst & Young LLP — EY India | May 2021 – May 2024 | Bangalore, India

- Developed ML/NLP solutions for sanctions screening, transaction monitoring, and conversational banking integrated into automation pipelines.
- Implemented hybrid entity matching, supervised scoring, and rule-based automation with Fircosoft integration while guiding junior developers.

Associate | City Union Bank Ltd | January 2018 – August 2019 | Chennai, India

- Handled retail banking operations, customer verification, transaction reconciliation, and core banking validation.
- Validated banking system entries and ensured regulatory compliance for customer transactions.

KEY PROJECTS

1. Mock Call Simulator — Insurance Employee Training | EY GDS

- **Objective:** Evaluate insurance product knowledge, customer handling, and empathy through realistic simulated calls.
- **Responsibilities:** Developed approximately 50 dynamic personas and configurable difficulty levels covering three topics each. Implemented LangGraph call-state management, optimized prompts to reduce redundant LLM calls, and integrated Azure Speech Services with RAG and agent workflows. Both phases deployed on AWS, supporting evaluations for 1,000+ employees.

- **Tools:** Python, FastAPI, LangGraph, LangChain, OpenAI GPT models, Azure Speech Services, AWS, GitLab, CloudBees.

2. AI for BI — Business Intelligence Assistant | EY GDS

- **Objective:** Enable conversational access to insurance documents, SharePoint files, charts, dashboards, and transactional data.
- **Responsibilities:** Built LangGraph/LangChain RAG and agent modules over thousands of documents, with APScheduler-based index refresh and schema-checked SQL retrieval. Regression-tested prompts and answers, delivered responses in 5–20 seconds depending on query complexity, and completed two-phase AWS delivery and client handover.
- **Tools:** Python, LangGraph, LangChain, OpenAI GPT-4, APScheduler, AWS, S3, SharePoint, PostgreSQL, GitLab, CloudBees.

3. Enterprise Finance & Operations Chatbot | Qentelli — Big Four Client

- **Objective:** Provide a unified interface for service-line financial reporting, budgets, forecasts, and P&L.
- **Responsibilities:** Built the platform from scratch and coordinated data access across Finance, Marketing, Operations, Infrastructure, and HR. Implemented multi-agent delegation, workflow state, Azure AI Search retrieval, structured tool outputs, and schema validation, with guardrails and exception escalation through Azure integrations.
- **Tools:** Python, LangGraph, LangChain, Azure OpenAI, Azure AI Foundry, Azure AI Search, Groq API, Azure Bot Service, Azure Functions, Azure Logic Apps.

4. Invoice Processing & Inventory Intelligence | Qentelli — Fastener Domain

- **Objective:** Automate invoice extraction, product classification, and quotation-to-inventory matching.
- **Responsibilities:** Built multi-format document processing using Tesseract OCR and NLP. Mapped products to catalogue master data, matched quotations against inventory for availability and estimates, and automated ingestion and processing through Dynamics 365 and Power Automate.
- **Tools:** Python, Tesseract OCR, NLP, Microsoft Dynamics 365, Power Automate, SQL.

5. Health Analytics & Interpretation Dashboard | Qentelli

- **Objective:** Convert medical documents and readings into health scores, trends, and conversational insights.
- **Responsibilities:** Built OCR/NLP/LLM processing workflows, ML-based health scoring and anomaly detection, and visualization dashboards. Integrated an Azure Bot Service assistant to provide conversational access to processed health information.
- **Tools:** Python, OpenAI GPT, LangChain, OCR, NLP, Azure Bot Service, Power BI, Tableau.

6. AML Sanctions & Name Screening Engine | EY India

- **Objective:** Identify high-risk watchlist matches and automate screening while retaining manual review for cases requiring investigation.
- **Responsibilities:** Built hybrid fuzzy, semantic, and phonetic matching with supervised ML, rule-based automation, and Fircosoft integration. Supported screening across millions of transactions, reducing false-positive rates from 55% to 10%. Automated processing previously requiring approximately three days of team-based manual review, completing it in three hours and routing only 2% of screened transactions for manual review.
- **Tools:** Python, SQL, Scikit-learn, spaCy, Fircosoft REST APIs, Power BI, AWS.

CERTIFICATIONS

- **Microsoft Certified:** Azure AI Engineer Associate — AI-102
- **Microsoft Certified:** Azure Fundamentals — AZ-900
- **GitHub Copilot Certification**

EDUCATION

Great Lakes Institute of Management

PGPM — Data Science and Engineering | September 2019 – March 2020

SASTRA University

B.Tech — Biotechnology | June 2013 – May 2017